Mst Afsana Mimi

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Research Areas

Statistical Machine Learning, Agentic AI, Relational Learning, Data Mining, and High-Dimensional Data Analysis.

Education

University of Massachusetts Amherst, Master's in Statistics (CGPA: 3.77) Sep 2023 – May 2026

- **Advisor:** Lulu Kang
- **Project: Gaussian Process Regression with Fairness Constraint** The project focuses on incorporating fairness constraints into Gaussian Process models that handle both qualitative and quantitative variables to reduce bias in predictions. It aims to ensure fair and reliable performance across protected groups while maintaining predictive accuracy.

University of Dhaka, Master's in Applied Statistics (CGPA: 3.75) Jan 2017 – Dec 2018

- **Advisor:** Md Hasinur Rahaman Khan
- **Project: Variable Selection for Censored Data Using Modified Correlation Adjusted Correlation (MCAR) Scores.** Developed an MCAR feature-selection method for censored high-dimensional data using AFT models. Evaluated predictive performance through simulations and DLBCL microarray data.

University of Dhaka, Bachelor's in Applied Statistics (CGPA: 3.61) Jan 2013 – Dec 2016

- **Advisor:** Md. Shafiqur Rahman
- **Project: Comparison Between Maximum Likelihood and Penalized Methods for Logistic Regression in Small Data.** Compared MLE and Firth's penalized logistic regression through simulation studies. Evaluated bias, MSE, and confidence interval performance, demonstrating Firth's advantages in small samples.

Experience

Research Assistant, Data Systems Group, DREAM Lab, Manning College of Information and Computer Sciences, University of Massachusetts Amherst Sep 2026 – Present

- **Advisor:** Marco Serafini
- **Project: Relational Predictive Query Processing.** Developing an agentic machine-learning framework for relational databases to automate schema understanding, relational evidence construction, and predictive reasoning over structured data. The long-term objective is to build a benchmark for evaluating agentic systems on realistic database tasks.

Adjunct Faculty, Department of Psychological and Brain Sciences, University of Massachusetts Amherst Sep 2026 – Present

Teaching Assistant, Department of Mathematics and Statistics, University of Massachusetts Amherst Sep 2023 – May 2026

Research Assistant, Institute of Statistical Research and Training, University of Dhaka Jul 2018 – Jun 2019

- **Project:** Poverty and Inequality Mapping in Bangladesh.

Technical Skills

Programming Languages: Python, R, C, SQL
Frameworks & Libraries: PyTorch, Torch Geometry
Statistical Software: SPSS, Stata, SAS, Tableau

Awards and Scholarship

- **Conference Award for Scientists (CASc)**, International Society for Clinical Biostatistics (ISCB), 2019.
- **National Science and Technology (NST) Fellowship Award** by Education Board of Bangladesh for MS thesis.

Publications

- Mimi, A., & Khan, M. H. R. (2021). Variable selection for censored data using modified correlation adjusted correlation (mcar) scores. Statistics in Medicine, 40(23), 5046–5064.

Google Scholar: https://scholar.google.com/citations?user=LaVyjwYAAAAJ&hl=en&oi=sra